

# Algorithm 1: Unified State Update Procedure

## Input

$\rho, \alpha, t$



## Phase 1

C2: Timeout  
Enforcement



## Phase 2

C1: Risk-Based  
Escalation



## Phase 3

C4: Human  
Approval Gate



## Phase 4

C5: Controlled  
De-escalation



## Phase 5

C3: Audit  
Logging



## Output

$s', \log \text{ entry}$